

Topic :- AI-Powered Smart Water Quality Monitoring System

Problem Statement:

Contaminated water is a leading cause of diseases worldwide. Traditional monitoring methods are slow, reactive, and inefficient, making it difficult to prevent waterborne illnesses. There is a need for an AI-driven system that can not only monitor water quality in real time but also predict potential diseases based on past contamination data.

Solution: AI-Powered Smart Water Quality Monitoring System with Disease Prediction

This system integrates IoT sensors, artificial intelligence, and cloud computing to continuously monitor water quality and predict potential health risks.

How It Works:

1. IoT Sensors Deployment – Sensors collect real-time data on pH, turbidity, dissolved oxygen, temperature, heavy metals, and bacterial contamination.
 2. AI-Powered Data Analysis – AI processes real-time and historical water quality data to detect contamination trends.
 3. Disease Prediction Model – Based on past data, the system maps water quality parameters to potential diseases and provides alerts to users.
 4. User Dashboard & Mobile App – Consumers, government agencies, and industries can see real-time water quality reports and view possible diseases linked to detected contamination.
 5. Automated Alerts & Preventive Actions – The system warns users if water is unsafe and suggests actions like boiling water, using filters, or reporting contamination.
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Dashboard Features: Real-Time Water Quality & Disease Prediction

The user-friendly dashboard will display:

- Real-Time Water Quality Parameters
 - pH Level

- Turbidity
 - Dissolved Oxygen
 - Conductivity (TDS)
 - Heavy Metal Contaminants (Lead, Arsenic, Mercury)
 - Bacterial Presence (E. coli, Cholera, Salmonella)
- AI-Powered Disease Prediction (Based on past data)
- If contamination is detected, the system will show:
- Possible Diseases: A list of illnesses that could arise from current water conditions.
 - Risk Level: Low, Moderate, or High (based on contamination severity).
 - Preventive Actions: Recommendations (e.g., use a water purifier, boil water, or avoid consumption).
- Alerts & Notifications
- Mobile & Email Alerts if contamination is found.
 - Government & Health Authorities Notified for critical contamination levels.

Example: How the Dashboard Works

Parameter Detected	Risk Level	Possible Diseases	Recommended Action
High Turbidity	Moderate	Gastroenteritis, Diarrhea	Use a water filter
Presence of E. coli Bacteria	High	Typhoid, Cholera, Dysentery	Boil water before use
Excess Nitrate (from runoff)	Moderate	Blue Baby Syndrome (Infants)	Avoid consumption
High Lead (Pb) Levels	High	Neurological disorders, Kidney damage	Use certified filters

Parameter Detected	Risk Level	Possible Diseases	Recommended Action
pH Imbalance (Too acidic)	Moderate	Skin Irritation, Stomach Problems	Neutralize with treatment

Benefits of AI-Powered Disease Prediction:

- Early Disease Prevention – Users are warned in advance about potential illnesses.
- Real-Time Decision Making – Consumers and authorities can take immediate action to ensure water safety.
- AI-Based Insights – Past data trends help predict future contamination risks.
- Healthier Communities – Reduces outbreaks of cholera, typhoid, and other waterborne diseases.
- Scalable & Smart – Can be used for municipal water supplies, industries, and homes.

This system ensures clean water, public health safety, and environmental protection, making it a game-changer in water quality monitoring.

Hardware Components:-

- ESP32
- pH Sensor – Measures water acidity
- TDS Sensor – Detects dissolved solids (water purity)
- Temperature Sensor (DS18B20) – Monitors water temperature
- Turbidity Sensor – Detects water clarity
- Blynk/Firebase – IoT dashboard for monitoring

Procedure:

1. Sensors collect data – pH, TDS, Turbidity, Temperature.
2. ESP32 processes & analyzes data – Compares values with safe limits.
3. Data is displayed on an OLED screen and sent to a mobile app (Blynk/Firebase).
4. If water is unsafe, a buzzer & LED alert users.

5. Cloud Dashboard shows real-time water quality trends.
6. AI Prediction– Can analyze past data to predict contamination risks.

Circuit Diagram :-

